Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-8642
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	**React.js / HTML5, CSS3, JavaScript**	Chosen for building an interactive user interface with drag-and-drop file upload, real-time import progress bars, and a visual field-mapping canvas.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	**Python (Django / FastAPI) or Node.js**	Selected for fast spreadsheet (Excel/CSV) parsing, processing transform scripts, managing coalesce (matching key) logic, and executing background batch jobs efficiently.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	**PostgreSQL / MySQL**	Used for managing temporary staging tables (Import Sets), mapping target application tables, and storing execution logs and error histories with strong data integrity.
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	**AWS (Amazon S3, EC2) & Docker**	AWS S3 provides secure cloud storage for uploaded spreadsheet files, while Docker containerization ensures seamless scalability and consistent deployment environments.
5	**Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)*	**GitHub & GitHub Actions**	Enables team source code collaboration, issue tracking, and automated build and test pipelines before deploying updates.

6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	**OpenPyXL / Pandas & Apache POI**	Provides robust spreadsheet parsing libraries to read .xlsx and .csv files, validate column headers, and execute automated field-mapping recommendations.